

# Kevin Berry

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## SKILLS

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Python, Rust, Go, C, JavaScript, Linux Kernel, Git, Machine Learning, Statistics, Containers

## EXPERIENCE

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### Software Engineer - [Container Optimized OS \(COS\)](#) Kernel

Jan 2024 - Present

*Google*

*Sunnyvale, CA*

- Led the development and maintenance of the kernel for an open-source Linux distro used as the basis of critical Google Cloud products like GKE and CloudSQL
- Worked with many teams to coordinate driver development and release efforts
- Tested drivers and integrated them into the COS kernel (Lustre, iRDMA, TPU, and others)
- Developed a package manager for kernel modules ([cos-dkms](#))
- Led investigations which resulted in order-of-magnitude speedups for file transfer workloads on COS
- Triaged and solved customer issues relating to many major Linux subsystems, e.g., networking, storage, etc.
- Triaged and patched kernel and package vulnerabilities

### Machine Learning Engineer

Jun 2017 - Dec 2023

*Worthix - Research Team*

*Alpharetta, GA*

- Developed LLM-based API for large scale multi-level feedback analysis and summarization
- Developed and trained NLP models and API for real-time survey response topic classification
- Developed novel algorithms for large-scale unsupervised graph classification
- Developed queries and statistical analyses for client-facing dashboards
- Deployed and monitored APIs and applications as scalable microservices in Azure
- Conducted scientific experiments to improve survey data and model quality
- Developed fast Monte-Carlo Shapley value approximation for calculating response topic importance
- Developed reverse-geolocation API capable of resolving millions of point-in-region queries per second
- Contributed Python code to networkx which massively improved performance for graph unions and intersections
- Contributed Rust code to polars dataframe library which solved infinite loop in core groupby operations

### Teaching Assistant

May 2016 - Dec 2017

*Computer Organization and Programming (CS 2110)*

*Atlanta, GA*

- Led recitations on C, Assembly, CPU datapaths, and digital logic
- Wrote software to automate grading of Java programs and circuits

## SELECTED PROJECTS

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**image-to-ascii (Rust):** Uses computer vision techniques to convert images and gifs to ASCII art. Capable of converting 120+ images per second. Uses hand-coded SIMD instructions for improved performance. ([Github](#)) ([Crates.io](#))

**detect-duplicates (Rust):** Finds all of the duplicated files in a directory. Reads each file at most twice and, in expectation, only has one file loaded at a time. Can check the entire Linux kernel tree in about 5 seconds. ([Github](#)) ([Crates.io](#))

**Readability Analyzer (JavaScript):** Assigns average text grade level based on standard readability metrics. Uses MLP with hand-engineered features to count syllables. Can analyze 2,000,000+ characters per second. ([Web App](#))

## EDUCATION

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Georgia Institute of Technology, B.S. Computer Science

Aug 2015 - Dec 2018

**Selected Coursework:** Machine Learning, Linguistics, Computer Graphics, Computer Vision, Compilers, Linear Algebra

**GPA:** 3.96/4.0